

Introduction to Information and Communication Technologies

CS-353 — Complete Lecture Notes

Part 1 & Part 2 | All Topics Covered

For First-Semester University Students

Pre-Engineering Background | Zero CS Experience Required

Credit Hours: 3 (3-0) | Pre-requisites: None

Topics Covered: 30 | Pages: 60+

Reference Books

Understanding Computers: Today & Tomorrow — Charles S. Parker
Computer Fundamentals — Pradeep K. Sinha & Priti Sinha
Computer Fundamentals — Anita Goel

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PART 1 — Foundations of Computing

Topics 1 – 16

✓ 1. Brief History of Computer

■ Definition

(A computer is an electronic device that can receive data (input), process it using instructions (a program), and produce a result (output).) The history of computers is the fascinating story of how human beings — over thousands of years — invented and improved machines to help them count, calculate, and process information, starting from simple counting beads and eventually reaching today's powerful microchip-based devices.

■ Detailed Explanation

To understand where computers come from, we need to travel far back in time. Long before electricity was discovered, people needed tools to count and calculate. (Ancient civilisations used the Abacus — a simple wooden frame with beads on rows of wires. By moving beads back and forth, people could add and subtract numbers. This was the world's first calculating tool, used thousands of years ago in China, Egypt, and Greece.)

(In 1642, a young French mathematician named Blaise Pascal built a mechanical calculator called the Pascaline. It was a small box with wheels and gears — turning one gear would turn the next, allowing simple addition and subtraction automatically. This was remarkable for its time because it was the first machine to do arithmetic without direct human effort for each step.

In 1671, German mathematician Gottfried Wilhelm Leibniz improved Pascal's design and built a machine that could also multiply and divide. He also promoted the binary number system (using only 0s and 1s) — the very foundation of how all modern computers store and process data.

The biggest conceptual leap came in the 19th century. Charles Babbage, an English mathematician, is called the 'Father of the Computer.' Around 1837, he designed the Analytical Engine — a steam-powered mechanical machine that had all the logical components of a modern computer: an input mechanism, a memory store, a processing unit, and an output mechanism. Although he never fully built it due to lack of money, his ideas were 100 years ahead of his time.)

Ada Lovelace, who worked with Babbage, is considered the world's first programmer. She wrote a detailed set of instructions for the Analytical Engine and understood that such a machine could perform any logical task, not just numbers.

In the early 20th century, things moved rapidly. In 1936, mathematician Alan Turing proposed the concept of a universal computing machine — a theoretical device that could simulate any computation. This became the foundation of computer science as we know it. By the 1940s, the first true electronic computers were built — enormous machines the size of entire rooms, using thousands of glass vacuum tubes. From there, the journey to today's smartphones took less than 80 years.

■ Real-Life Example

Think of it exactly like the evolution of transportation. Humans first walked everywhere. Then came horses, then the wheel, then horse-drawn carriages, then steam-powered trains, then automobiles, and finally airplanes. Each invention built on the last and was faster and more capable. The history of computers follows the same pattern — each era built on the discoveries of the one before.

■ Short Summary

Computers evolved over thousands of years — from simple counting tools like the abacus, to mechanical calculators by Pascal and Leibniz, to Babbage's conceptual Analytical Engine, and finally to electronic computers in the 1940s. Charles Babbage is the Father of the Computer. Ada Lovelace is considered the first programmer.

✓ 2. Four Stages of Computer History

■ Definition

The history of electronic computers is divided into four main stages, called Generations. Each generation is defined by a major change in the core technology used to build computers. Moving from one generation to the next, computers became smaller, faster, cheaper, more reliable, and more powerful.

■ Detailed Explanation

First Generation — 1940s to 1950s (Vacuum Tubes)

The first electronic computers used Vacuum Tubes — glass bulbs from which air had been removed, containing metal elements that could control electrical signals. These tubes acted as switches, turning electric current on and off to represent binary 0s and 1s.

Computers of this era were enormous — sometimes filling entire rooms or even entire buildings. They weighed several tons, consumed huge amounts of electricity, generated extreme heat, and broke down very frequently because the tubes burned out constantly. Programming was done in Machine Language — long strings of 0s and 1s typed directly. Famous examples include ENIAC (Electronic Numerical Integrator and Computer) and UNIVAC, the first commercially sold computer.

Second Generation — 1950s to 1960s (Transistors)

Transistors replaced vacuum tubes. A transistor is a tiny semiconductor device that does the same job as a vacuum tube (switching and amplifying electrical signals) but is far smaller, uses much less power,

produces far less heat, is much more reliable, and lasts much longer.

With transistors, computers became significantly smaller, cheaper, and more dependable. Programming moved forward too — Assembly Language was introduced, which used short codes (like ADD, SUB) instead of raw 0s and 1s, making programming somewhat easier.

Third Generation — 1960s to 1970s (Integrated Circuits)

The breakthrough of this era was the Integrated Circuit (IC) — a single small silicon chip that could contain dozens, then hundreds, then thousands of transistors. Instead of connecting individual components with wires, everything was etched onto one tiny chip.

This made computers dramatically smaller, faster, and more affordable. Multiple programs could run at the same time (multiprogramming). High-level programming languages like COBOL, FORTRAN, and BASIC were developed — languages that looked almost like English, making programming accessible to far more people. Keyboards and monitors replaced punch cards and paper printouts as the main way to interact with the computer.

Fourth Generation — 1970s to Present (Microprocessors)

The defining invention of this generation is the Microprocessor — an entire Central Processing Unit (CPU) contained on a single tiny chip. Intel introduced the first commercial microprocessor, the Intel 4004, in 1971.

Microprocessors made Personal Computers (PCs) possible — computers small and cheap enough for individuals to own at home. The Apple II (1977) and the IBM PC (1981) brought computing to ordinary people. This generation also gave us graphical user interfaces (clicking icons instead of typing commands), the internet, smartphones, laptops, tablets, and now artificial intelligence. We are still in the Fourth Generation today.

■ Real-Life Example

Think of televisions. The first TVs were enormous wooden cabinets showing black-and-white pictures with poor reception (like first-generation computers). Then came smaller colour TVs (second generation). Then slimmer, sharper flat-screen TVs (third generation). Now we have thin Smart TVs with 4K resolution, streaming, and voice control (fourth generation). Each era was a dramatic improvement — just like the four generations of computers.

■ Short Summary

The four generations of computers: 1st used vacuum tubes (large, hot, slow); 2nd used transistors (smaller, reliable); 3rd used integrated circuits (very small, multiprogramming, high-level languages); 4th uses microprocessors (personal computers, internet, smartphones). Each generation brought smaller size, faster speed, lower cost, and greater capability.

3. Computer Elements — An Overview

■ Definition

A computer system is made up of several key elements (components) that work together to receive information, process it, store it, and deliver a result. The four fundamental elements are: Input, Processing, Storage, and Output. These form what is known as the IPO Cycle — the basic working model of every computer.

■ Detailed Explanation

No matter how simple or how powerful a computer is — whether it is a tiny calculator or a massive supercomputer — it always follows the same four-step cycle. Understanding this cycle is the key to understanding how computers work.

Input

Input is the information or data that you provide to the computer. This can come in many forms: text typed on a keyboard, a click of a mouse, a photo taken by a camera, a voice recorded by a microphone, or data read by a barcode scanner. Without input, the computer has nothing to work with.

Processing

Processing is what the computer does with the input — it follows the instructions in a program to calculate, compare, sort, or transform the data. This work is done by the Central Processing Unit (CPU), which is the brain of the computer. Processing happens at incredible speed — billions of operations per second.

Storage

Storage is where the computer keeps data — either temporarily while it is working (in RAM) or permanently for future use (on a hard drive or USB drive). Without storage, every piece of data would be lost the moment the power was cut.

Output

Output is the result of the processing — what the computer delivers back to the user. This could be text displayed on a monitor, a document printed by a printer, a sound played through speakers, or data sent to another device. Output is the purpose of the entire process.

These four elements form a continuous cycle. Storage supports all three other elements — the CPU reads instructions from storage, works on data in storage, and saves results back to storage.

■ Real-Life Example

When you search something on Google (Input: you type a query), Google's servers process your request and look for matching pages (Processing), retrieve information from their massive databases (Storage), and display the results on your screen (Output). This entire process happens in less than one second.

■ Short Summary

Every computer works on the IPO Cycle: Input (data given to the computer), Processing (CPU works on it), Storage (data saved temporarily or permanently), and Output (result delivered to the user). All four elements must work together for the computer to function.

4. The Processor (CPU)

■ Definition

The Central Processing Unit (CPU) is the main chip inside a computer that carries out all instructions and performs all calculations. It is the brain of the computer — every single action the computer takes is directed by the CPU. Without a CPU, a computer simply cannot function.

■ Detailed Explanation

Think of the CPU as the manager of a very busy office. Every request that comes in — from opening a file, to playing a video, to calculating a sum — goes to the CPU. The CPU reads the instructions, decides what needs to be done, does the work, and sends the result to the right place.

Parts of the CPU

The CPU has two main internal parts. The Arithmetic Logic Unit (ALU) performs all arithmetic operations (addition, subtraction, multiplication, division) and all logical operations (comparing values — is A equal to B? Is X greater than Y?). The Control Unit (CU) is the organiser — it fetches instructions from memory, decodes them (figures out what they mean), and directs the ALU and other components to execute them. It does not do calculations itself but coordinates everything else.

CPU Speed and Cores

CPU speed is measured in GHz (GigaHertz). One GHz means the CPU completes one billion instruction cycles per second. A modern laptop CPU might run at 2.5 to 4.5 GHz — meaning it processes billions of operations every single second.

Modern CPUs have multiple cores — think of each core as an additional brain working in parallel. A dual-core CPU can handle two streams of instructions simultaneously. A quad-core handles four. High-end CPUs today have 8, 12, or even 16 cores, allowing them to run many programs at the same time without slowing down.

■ Real-Life Example

Imagine a very fast accountant sitting inside your computer. Every time you click something — open a program, save a file, play music — that accountant instantly reads the request, does all the necessary calculations, and tells the screen what to show or tells the hard drive what to save. That accountant never takes a break and works billions of times per second. That is the CPU.

■ Short Summary

The CPU is the brain of the computer. It has two main parts: the ALU (performs calculations and logic) and the Control Unit (fetches, decodes, and directs instructions). Speed is measured in GHz. Modern CPUs have multiple cores to handle many tasks at the same time.

5. Memory

■ Definition

Memory in a computer is the hardware that stores data and instructions — either for immediate use while the computer is working, or for long-term keeping. The two broad categories are Primary Memory (used directly by the CPU) and Secondary Memory (used for permanent long-term storage).

■ Detailed Explanation

Primary Memory — RAM and ROM

Primary Memory is the memory that the CPU directly accesses to run programs. It is fast but limited in size. RAM (Random Access Memory) is the most important type. It temporarily holds the programs and data you are currently working with. It is called 'volatile' memory — when you turn off the computer, everything in RAM disappears. The more RAM a computer has, the more programs it can run smoothly at the same time.

ROM (Read-Only Memory) is a small chip that permanently stores the computer's basic startup instructions — called the BIOS (Basic Input/Output System). When you turn on your computer, the CPU reads from ROM first to know how to start. ROM is 'non-volatile' — its contents are not lost when power is removed, and you cannot normally change what is written in it.

Secondary Memory

Secondary Memory provides large, permanent storage. Examples include Hard Disk Drives (HDD), Solid State Drives (SSD), USB flash drives, CDs, DVDs, and cloud storage. Unlike RAM, data in secondary memory survives even when the computer is off. This is where your documents, photos, videos, and programs are permanently saved.

Units of Memory

Memory capacity is measured in bytes. One byte stores one character. Common units: 1 Kilobyte (KB) = 1,024 bytes. 1 Megabyte (MB) = 1,024 KB. 1 Gigabyte (GB) = 1,024 MB. 1 Terabyte (TB) = 1,024 GB. A typical modern laptop might have 8 GB or 16 GB of RAM and 512 GB or 1 TB of storage.

■ Real-Life Example

Think of RAM as your actual study desk — you can only place a limited number of books (open programs) on it at once. Your hard drive is the large bookshelf on the wall — it holds all your books permanently. When you want to study from a book, you take it from the shelf and put it on your desk. When you close it, it goes back to the shelf. RAM is the desk; hard drive is the shelf.

■ Short Summary

Memory stores data for the computer. Primary Memory: RAM (temporary, volatile, used while working) and ROM (permanent, startup instructions). Secondary Memory: hard drives, USB drives (permanent, large capacity). Memory size is measured in bytes, KB, MB, GB, and TB.

6. Hardware

■ Definition

Hardware refers to all the physical, tangible components of a computer system — the parts you can see, touch, and feel. This includes everything from the CPU chip inside the computer, to the keyboard you type on, to the screen you look at, and the printer you use to print documents.

■ Detailed Explanation

Hardware is the physical body of the computer. You can divide it into two broad groups.

Internal Hardware

These are the components located inside the computer case. The Motherboard is the main circuit board — it connects all components together and allows them to communicate. The CPU (processor) is the brain. RAM chips are the temporary memory. The Hard Drive stores data permanently. The Power Supply Unit (PSU) converts electricity from the wall socket into the voltages the computer needs. The Graphics Card (GPU) handles visual output to the screen. The Cooling Fan prevents the CPU from overheating.

External Hardware (Peripherals)

These are devices connected from the outside of the computer. Input peripherals include the keyboard, mouse, scanner, microphone, and webcam. Output peripherals include the monitor, printer, and speakers. Storage peripherals include USB flash drives and external hard drives.

Hardware and software must always work together. Hardware provides the physical capability; software provides the instructions. Without software, hardware is just expensive metal and plastic that does nothing. Without hardware, software has no machine to run on.

■ Real-Life Example

Think of the human body. Your brain, heart, hands, eyes, and ears are all hardware — physical organs that exist and can be touched. The knowledge, skills, language, and habits stored in your brain are the software — they are not physical but they direct how the hardware (your body) behaves. A computer works the same way.

■ Short Summary

Hardware is every physical component of a computer. Internal hardware (inside the case): CPU, RAM, motherboard, hard drive, power supply. External hardware (peripherals): keyboard, mouse, monitor, printer. Hardware and software always work together — neither is useful without the other.

7. Software

■ Definition

Software is a collection of programs, instructions, and data that tell the computer's hardware what to do. Unlike hardware, software has no physical form — you cannot touch it. It exists as digital code stored in memory or on a disk. Without software, hardware is completely useless. Software is divided into two main types: System Software and Application Software.

■ Detailed Explanation

If hardware is the body of the computer, software is its mind. Software tells every piece of hardware exactly what to do, when to do it, and how to do it.

When you press the power button on your laptop, the operating system (a type of software) loads and starts managing the CPU, memory, and all connected devices. When you open a web browser, that is another piece of software running. When you play a game, watch a video, or write a document — in every case, software is what makes it happen.

System Software

System software manages the hardware and provides a base for application software to work on. The most important example is the Operating System (OS) — such as Windows, macOS, Linux, or Android. System software also includes device drivers (which let the OS communicate with hardware like printers) and utility programs (like antivirus tools and disk cleaners).

Application Software

Application software is designed for specific tasks that the end user wants to perform. Examples include MS Word for writing, MS Excel for spreadsheets, Photoshop for editing images, Chrome for browsing the internet, and VLC for playing videos.

Software is created by programmers who write code in programming languages like Python, Java, C++, and JavaScript. Once written, the software is compiled into a form the computer can execute and stored on a disk, ready to be loaded into RAM when needed.

■ Real-Life Example

Imagine a piano. The piano itself — the keys, strings, hammers, and wooden frame — is the hardware. The sheet music that tells the pianist which keys to press and in what order is the software. Without sheet music (instructions), the piano sits silent. Without the piano (hardware), the sheet music is just marks on paper. Both are needed to make music.

■ Short Summary

Software is coded instructions that tell hardware what to do. It has no physical form. Two main types: System Software (manages the computer — e.g., Windows OS) and Application Software (performs specific user tasks — e.g., MS Word, Chrome). Programmers write software using programming languages.

8. Application Software — Uses and Limitations

■ Definition

Application Software (commonly called 'apps' or 'programs') is software designed to help the user perform specific, real-world tasks. It runs on top of the operating system and is the software that most users interact with every single day — from writing documents to playing games to sending emails.

■ Detailed Explanation

Categories of Application Software

Word Processing Software such as Microsoft Word and Google Docs is used to write, edit, format, and print text documents — reports, letters, essays, and books. Spreadsheet Software such as Microsoft Excel is used to organise, calculate, and analyse numerical data, and to create charts and graphs. Presentation Software such as Microsoft PowerPoint is used to create visual slide-based presentations for meetings or lectures. Database Software such as Microsoft Access organises and retrieves large amounts of structured data. Web Browsers such as Google Chrome and Mozilla Firefox allow users to access and navigate the internet. Media Players such as VLC play audio and video files. Graphics and Design Software such as Adobe Photoshop and Canva are used for image editing and graphic design. Communication Apps such as WhatsApp, Gmail, and Zoom are used for messaging, email, and video calls.

Uses of Application Software

Application software makes complex tasks simple and fast. It is used in virtually every field: in business for accounting, sales, and customer management; in education for online learning and research; in healthcare for managing patient records and diagnostic tools; in entertainment for streaming music, videos, and gaming; in engineering for design and simulation; and in communication for staying connected with others globally.

Limitations of Application Software

Each application is built for a specific purpose and generally cannot perform tasks outside that purpose — you cannot edit photos in Microsoft Word. Some professional software is very expensive to purchase or subscribe to. Not all software is compatible with every operating system (a Windows application may not work on macOS). Software can contain bugs (coding errors) that cause crashes or incorrect results. Applications require minimum hardware specifications (enough RAM and CPU speed) to run properly — running heavy software on an old computer can cause it to freeze or crash.

■ Real-Life Example

Microsoft Excel is application software built specifically for working with numbers and data. It can sum columns, create pie charts, apply formulas, and manage budgets beautifully. However, it cannot browse the internet, play music, or edit a photograph — those tasks require different applications. This specialisation is both the strength and the limitation of application software.

■ Short Summary

Application software helps users perform specific tasks: writing (Word), calculating (Excel), browsing (Chrome), designing (Photoshop), communicating (WhatsApp). Its main limitation is that each app is designed for one purpose and cannot go beyond it. Some are expensive, have compatibility issues, and require sufficient hardware to run.

9. System Software — Importance and Types

■ Definition

System Software is the software that manages and controls the computer's hardware and provides a foundation upon which application software can run. It works largely in the background — invisible to most users — ensuring that all hardware components function correctly and that applications have the environment they need to operate.

■ Detailed Explanation

Operating System (OS)

The Operating System is the most important piece of system software. It is the first software loaded when the computer powers on, and it remains active until the computer is shut down. The OS manages the CPU (deciding which program gets processor time), manages memory (allocating RAM to running programs), manages storage (organising files and folders on the disk), manages input/output devices (communicating with the keyboard, mouse, screen, and printer), and provides the user interface — the visual environment you see and interact with.

Common operating systems include Windows (by Microsoft, used on most PCs), macOS (by Apple, used on Mac computers), Linux (free and open-source, used on servers and by programmers), Android (by Google, used on most smartphones), and iOS (by Apple, used on iPhones).

Device Drivers

A device driver is a small piece of system software that acts as a translator between the operating system and a specific hardware device — like a printer, graphics card, or sound card. When you plug in a new printer, the OS uses the printer's driver to understand how to communicate with it. Without the correct driver, the hardware device will not work, even if it is physically connected.

Utility Software

Utility programs are system software tools that help maintain, repair, and optimise the computer. Common examples include: Antivirus software (detects and removes malicious programs), Disk Cleanup tools (remove unnecessary files to free up space), File Compression tools like WinZip (reduce file sizes for easier storage and sharing), Backup software (automatically copies important data to a safe location), and Defragmentation tools (reorganise data on a hard disk to improve speed).

■ Real-Life Example

Imagine building a new shopping mall (the computer). The mall needs a management company (the operating system) to control electricity, air conditioning, security, and decide which shops can open and where. Each shop (application software) needs a specific permit and setup from the management. The maintenance crew (utility software) keeps the building clean and functional. Without management and maintenance, the mall cannot operate — and the same is true for a computer without system software.

■ Short Summary

System software manages the computer. Types: Operating System (manages hardware, provides user interface — Windows, macOS, Android), Device Drivers (translate between OS and hardware), and Utility Software (maintains and optimises the computer — antivirus, disk cleaner). Without system software, no computer can function.

10. Types of Computers

■ Definition

Computers exist in many different sizes and have very different levels of processing power, depending on the tasks they are designed to perform. The four main types are: Supercomputers (the most powerful), Mainframe Computers (large business machines), Minicomputers (medium-sized), and Microcomputers (personal computers we use every day).

■ Detailed Explanation

Supercomputers

Supercomputers are the most powerful computers ever built by humans. They are designed for tasks requiring an astronomical number of calculations in a very short time — such as predicting weather patterns, simulating nuclear reactions, modelling climate change, running artificial intelligence systems, or decoding the human genome. A supercomputer can perform quadrillions (1,000,000,000,000,000) of calculations per second. They cost millions or even billions of dollars to build and operate. Famous examples include Frontier (the world's fastest supercomputer as of recent years, located in the USA) and Param Shivay (India). They require entire specially cooled buildings and teams of engineers to run.

Mainframe Computers

Mainframe computers are very large, powerful machines used by big organisations — such as banks, airlines, insurance companies, government departments, and large corporations. They can handle thousands of users simultaneously and process enormous volumes of data continuously. When you use an ATM machine, withdraw money from your bank account, or book an airline ticket, the transaction is very likely being processed by a mainframe computer somewhere. They are extremely reliable — designed to run continuously for years without shutting down.

Minicomputers (Midrange Computers)

Minicomputers are medium-sized computers, more powerful than a personal computer but less powerful than a mainframe. They are used by medium-sized businesses, research laboratories, universities, and

government offices for tasks like managing databases, running business applications, and supporting multiple users at the same time. They are significantly smaller and cheaper than mainframes. The term 'minicomputer' is less commonly used today because modern servers perform similar roles.

Microcomputers (Personal Computers)

Microcomputers — also called Personal Computers (PCs) — are the computers most familiar to us. These include desktop computers, laptops, tablets, and smartphones. They are designed to be used by one person at a time and are affordable enough for home and office use. They are used for everyday tasks: writing documents, browsing the internet, watching videos, playing games, video calls, and much more. The microprocessor chip is what makes microcomputers small, cheap, and powerful enough for personal use.

■ Real-Life Example

Think of vehicles as an analogy. A bicycle is like a microcomputer — small, personal, affordable, used by one person for daily needs. A car is like a minicomputer — slightly more powerful, suitable for a small group. A large truck or freight lorry is like a mainframe — built for heavy loads, used commercially. A rocket or space shuttle is like a supercomputer — the pinnacle of engineering, used only for the most extraordinary missions.

■ Short Summary

Four types of computers: Supercomputers (most powerful, scientific research, climate and AI simulations), Mainframes (large, handle thousands of users, used by banks and airlines), Minicomputers (medium-sized, for businesses and universities), and Microcomputers (personal computers — desktops, laptops, tablets, smartphones).

11. Introduction to CBIS (Computer-Based Information System)

■ Definition

A Computer-Based Information System (CBIS) is a system that uses computer hardware, software, and human expertise to collect, store, process, and communicate information within an organisation. Its purpose is to help organisations manage data efficiently and support better, faster decision-making at all levels.

■ Detailed Explanation

Every organisation — a hospital, a university, a bank, a government department, or a company — generates and needs to manage large amounts of information daily. Without a proper system, this information becomes impossible to manage accurately. CBIS uses computers to solve this problem.

Five Components of a CBIS

Hardware is the physical equipment — computers, servers, printers, and network devices. Software is the programs used to process and manage data — database management software, accounting programs, and reporting tools. Data is the raw information entered into the system — names, numbers, dates, and

records. Procedures are the rules, guidelines, and instructions for how the system is used by employees. People are the users who operate the system, enter data, manage it, and make decisions based on the outputs it produces. All five components must work in harmony for the CBIS to be effective.

Types of CBIS

A Transaction Processing System (TPS) handles the routine daily transactions of an organisation — recording sales, processing payments, updating stock levels, and managing payroll. A Management Information System (MIS) takes raw data from the TPS and produces summarised reports and charts that help middle managers make decisions — for example, a monthly sales report showing which products sold best. A Decision Support System (DSS) helps senior managers with complex, non-routine decisions by analysing data and modelling different scenarios — for example, helping a company decide whether to expand to a new city.

■ Real-Life Example

Imagine a large hospital. Doctors need access to patient medical records. Nurses need to see medication schedules. Administrators need to manage billing and appointments. A CBIS allows all of them to access the same centralised, up-to-date information from their own computers. If a patient is admitted, their record is immediately visible to every department that needs it. This prevents dangerous errors, saves time, and improves the quality of care.

■ Short Summary

CBIS uses computers to manage organisational information. It has five components: hardware, software, data, procedures, and people. Types include TPS (daily transactions), MIS (management reports), and DSS (complex decision support). CBIS improves accuracy, speed, and decision-making across the organisation.

12. Methods of Input and Processing

■ Definition

Methods of Input refer to the different techniques used to enter data into a computer system. Methods of Processing refer to how the computer handles and works on that data once it has been entered. Different situations, industries, and data types require different input and processing approaches.

■ Detailed Explanation

Methods of Input

Manual Data Entry is the most common method — a person types data using a keyboard or enters it through a form on screen. This is flexible but slow and prone to human error for large amounts of data. Batch Input involves collecting a large quantity of data first and then entering it all at once at a scheduled time — for example, processing all employee payroll data at the end of the month. Online (Real-Time) Input means data is entered immediately as events happen — like a cashier scanning items at a

supermarket checkout, or a receptionist recording a patient's details as they arrive at a clinic.

Source Data Automation (SDA) eliminates manual typing by reading data automatically from the original source. Examples include barcode scanners in supermarkets (reading product codes automatically), ATM card readers (reading your bank card), OCR (Optical Character Recognition — machines that read printed or handwritten text and convert it to digital text), OMR (Optical Mark Recognition — reading filled-in bubbles on exam answer sheets), and MICR (Magnetic Ink Character Recognition — reading the special numbers printed at the bottom of cheques).

Methods of Processing

Batch Processing means all collected data is processed together as a group at a specific scheduled time, not immediately. This is efficient for large volumes of similar data. Examples: generating electricity bills for thousands of households at the end of each month, or processing all exam results at the end of a semester. Real-Time (Online) Processing means data is processed the instant it is entered, and results are immediately available. Examples: online banking (your balance updates the second you transfer money), airline booking systems (a seat becomes unavailable the moment someone books it). Multi-programming allows the CPU to work on multiple programs at the same time by rapidly switching between them, so all programs appear to run simultaneously.

■ Real-Life Example

Think of a bakery. Manual input is a customer walking in and personally telling the baker what they want. Source data automation is an ordering kiosk where the customer scans a product code. Batch processing is the baker collecting all orders from the morning and baking everything together at noon in one big batch. Real-time processing is a fully automated baking machine that immediately starts making your item the moment your order is placed and confirmed.

■ Short Summary

Input methods: manual entry, batch input, online input, and Source Data Automation (barcode scanners, OCR, MICR). **Processing methods:** Batch Processing (process all data at once, scheduled) and Real-Time Processing (process immediately as data arrives). The appropriate method depends on the organisation's needs, data volume, and urgency of results.

13. Organizing a Computer Facility

■ Definition

Organizing a Computer Facility means deciding how an organisation will set up, manage, and distribute its computing resources — its computers, servers, data, and IT staff. The central question is: should all computing be controlled from one place, spread across multiple connected locations, or left entirely independent in each department?

■ Detailed Explanation

When a large organisation — a bank, a university, a hospital, a government body — uses computers, it must make a critical decision about how to organise its IT infrastructure. This choice affects everything: cost, speed, security, reliability, scalability, and how easily employees across different locations can access the data they need.

The three main models of organising a computer facility are: Centralised, Distributed, and Decentralised. Understanding the difference between these models is very important for anyone working in or managing an organisation.

Key factors that influence this decision include: How many users need simultaneous access? How critical is data security and centralised control? How fast must the system respond? Does each department need independence and customisation? What is the budget for IT infrastructure?

■ Real-Life Example

Think of a restaurant chain with 50 branches across the country. A centralised approach means all orders, menus, and accounts are managed from one central head office. A distributed approach means each branch has its own small kitchen and management, but all are connected to share data with head office. A decentralised approach means each branch is completely independent — it sets its own menu, manages its own accounts, and has no connection to the others. Each model has very different costs, advantages, and risks.

■ Short Summary

Organising a computer facility is about deciding how to set up and control computing resources across an organisation. The three models — Centralised, Distributed, and Decentralised — offer different trade-offs between control, cost, reliability, and flexibility. The right choice depends on the organisation's size, needs, and budget.

14. Centralised Computing Facility

■ Definition

In a Centralised Computing Facility, all computer hardware, software, data, and processing power are located in one single central location — usually a specially managed Data Centre. All users across the organisation, regardless of where they are physically located, connect to this one central computer or server to access data and run programs.

■ Detailed Explanation

In the centralised model, there is one powerful central computer (typically a mainframe or a large server). All employees — whether in the accounts department, the sales team, or a distant branch office — access this central system remotely through terminals or networked computers. The central system does all the processing; the terminals just display results and accept input.

Advantages

Easier to manage and maintain — everything is in one place, so the IT team only needs to manage one system. Data security is stronger — all data is stored and backed up in one controlled, secure location. Cost-effective — one powerful central server is often cheaper overall than buying and maintaining dozens of individual powerful computers. Consistency — all users access the same, most up-to-date version of data and software. Updates and changes are made once and are immediately available to everyone.

Disadvantages

Single Point of Failure — if the central computer breaks down, the entire organisation is paralysed. Nobody can work until the problem is fixed. Bottleneck — if too many users try to access the system at the same time, it can become slow. Distance issues — users who are physically far from the central computer may experience slower response times. It requires a very reliable, high-capacity central machine and a fast, stable network connecting all users to it.

■ Real-Life Example

Think of a university with one central library that holds all the books and resources. Every student and lecturer must use this one library. The books are well-organised, secure, and maintained by expert staff (advantages). But if the library closes — due to a fire, renovation, or power cut — nobody can access any resources at all (disadvantage).

■ Short Summary

Centralised Computing places all resources in one central location. Advantages: easy management, strong security, cost savings, data consistency. Disadvantages: single point of failure (entire system stops if central machine fails), potential bottlenecks with many users, and slow access for distant users.

15. Distributed Computing Facility

■ Definition

In a Distributed Computing Facility, computing resources — computers, servers, and data — are spread across multiple locations, all connected together through a network. Each location can process data locally (close to the user) but can also share data and resources with other connected locations across the network.

■ Detailed Explanation

In the distributed model, instead of one central all-powerful computer, there are multiple computers or servers located at different departments, branches, cities, or even countries — all linked together by a network. Each location has enough computing power to handle its own work independently, but the network allows data and resources to be shared across all locations.

Advantages

No Single Point of Failure — if one computer or one location goes down, the rest of the network continues to work normally. Local Processing Speed — since processing happens locally (at or near the user's location), responses are faster than in a centralised system where data has to travel to a distant central server. Flexibility and Scalability — new computers or new locations can be added to the network easily as the organisation grows. Redundancy — data can be replicated (copied) across multiple locations, so even if one site is destroyed, the data is safe at another.

Disadvantages

More Complex to Manage — with resources spread across many locations, the IT team faces a more complicated management task than with a centralised system. Data Consistency challenges — keeping all copies of data perfectly synchronised and up-to-date across all locations requires careful coordination. Security is more complex — there are more locations and network connections to protect. Initial setup cost is higher — establishing the network connections between all locations can be expensive.

■ Real-Life Example

A bank with branches in 20 cities is a perfect example of distributed computing. Each branch has its own servers and can process transactions locally, giving customers fast service. But all branches are connected — so if you open an account at the Karachi branch, a customer can access that same account at the Lahore branch immediately. If one branch's system crashes, the others keep running.

■ Short Summary

Distributed Computing spreads resources across multiple networked locations. Advantages: no single point of failure, faster local processing, easy scalability, and data redundancy. Disadvantages: complex management, data synchronisation challenges, and more complex security requirements.

16. Decentralised Computing Facility

■ Definition

In a Decentralised Computing Facility, each department, branch, or unit of an organisation manages its own computers, software, and data completely independently — with no central control and often little or no network connection between different units. Each unit is entirely self-sufficient and makes its own IT decisions.

■ Detailed Explanation

In the decentralised model, there is no head IT department, no central server, and no shared network. Each department purchases its own equipment, installs its own software, hires its own IT staff if needed, and manages its own data — entirely on its own terms.

Advantages

Maximum Autonomy — each department has full control over its own computing environment and can make decisions quickly without waiting for approval from a central authority. Customisation — each department can set up systems exactly tailored to its own specific needs and workflows. Isolation — if one department's system fails, it has absolutely no impact on other departments.

Disadvantages

Highest Cost — since every department must buy and maintain its own hardware and software separately, the total cost across the organisation is much higher than a centralised or distributed approach. Data Sharing Problems — sharing information between departments is extremely difficult. If the accounts department needs data from the sales department, they must physically transfer files or manually re-enter data — slow, error-prone, and inefficient. Duplication of Effort — multiple departments may end up doing the same work independently, wasting both time and money. No uniform standards — different departments may use incompatible software, making collaboration very difficult.

Comparison Summary

Centralised provides maximum control with the risk of total failure. Distributed balances local independence with shared connectivity. Decentralised provides maximum independence but the highest cost and the greatest coordination challenges.

■ Real-Life Example

Imagine a large university where the Faculty of Science, the Faculty of Arts, and the Faculty of Business each have their own completely separate computer labs, their own IT staff, their own software licenses, and their own data servers — with no connection between them at all. The Science faculty uses Linux, Arts uses Windows, and Business uses macOS. They cannot easily share files or databases. Each operates as a completely independent computing island.

■ Short Summary

Decentralised Computing gives each department or unit full independence to manage its own computers with no central control. Advantages: maximum autonomy, customisation, and isolation from other departments' failures. Disadvantages: highest cost, very difficult data sharing between departments, duplication of effort, and no uniform standards.

PART 2 — Devices, Storage, Networks & Internet

Topics 17 – 30

17. Input Devices — Keyboard and its Types

■ Definition

An input device is any hardware that allows a user to enter data or instructions into a computer. The keyboard is the most common and most important input device. It allows users to type text, numbers, symbols, and commands into the computer by pressing individual keys.

■ Detailed Explanation

Without input devices, there would be no way for a human to communicate with a computer. Input devices convert human actions — pressing a key, moving a mouse, speaking a word, scanning a barcode — into electronic signals that the computer can understand and process.

The Keyboard

The keyboard is made up of a set of keys arranged in a specific layout. Most keyboards follow the QWERTY layout (named after the first six letters on the top row), which was originally designed for typewriters. Each key, when pressed, sends an electronic signal to the computer representing that character or command.

A standard keyboard is divided into groups: Alphanumeric keys (letters A–Z and numbers 0–9 for typing text and numbers), Function keys (F1 through F12 along the top row, used as shortcuts in different programs), Numeric keypad (a separate block of number keys on the right, like a calculator), Control keys (Enter, Backspace, Delete, Shift, Ctrl, Alt — used to perform commands and control the cursor), Navigation keys (arrow keys and keys like Home, End, Page Up, Page Down — used to move around a document).

Types of Keyboards

A Standard Keyboard is the traditional full-size keyboard with all key groups, connected to the computer via a cable (USB) or wirelessly. A Mechanical Keyboard uses individual physical switches under each key that click and give tactile feedback — preferred by programmers and gamers for accuracy and feel. A Membrane Keyboard uses a soft pressure-sensitive membrane layer under the keys — quieter, cheaper, and thinner than mechanical keyboards. An Ergonomic Keyboard is specially designed with a split, curved layout to reduce strain on the wrists and hands during long typing sessions. A Wireless Keyboard connects

via Bluetooth or a USB receiver — removes the need for a physical cable, allowing more flexibility. A Virtual (On-Screen) Keyboard appears on a touchscreen and is operated by tapping the screen — used on smartphones, tablets, and some all-in-one PCs.

■ Real-Life Example

When you write an essay on your laptop, every letter, space, and full stop you produce is because you pressed a key on the keyboard. The keyboard translates your physical finger presses into electrical signals, which the operating system interprets and passes to MS Word, which then displays the character on screen. The entire process takes less than a millisecond.

■ Short Summary

A keyboard is the most common input device, used to type text and commands. Key groups include alphanumeric, function, numeric keypad, control, and navigation keys. Types of keyboards: Standard, Mechanical (physical switches), Membrane (soft layer, quiet), Ergonomic (reduces wrist strain), Wireless (no cable), and Virtual (on-screen touchscreen).

18. Terminals — Dumb, Smart, and Intelligent

■ Definition

A terminal is a device — usually consisting of a screen and a keyboard — that is used to interact with a central computer or server. The terminal itself may or may not have its own processing power. Based on their capability, terminals are classified into three types: Dumb Terminals, Smart Terminals, and Intelligent Terminals.

■ Detailed Explanation

In the early days of computing, most organisations had one large central computer shared by many users. Each user accessed this central computer from their own desk using a terminal — a screen-and-keyboard combination. The terminal was simply the window through which the user communicated with the powerful central machine.

Dumb Terminal

A Dumb Terminal has no processor and no memory of its own. It is simply a display screen and a keyboard. It cannot process any data on its own — it can only send keystrokes to the central computer and display whatever the central computer sends back. All the actual computing happens at the central machine. Dumb terminals are very cheap and simple. They were commonly used in banks, airline reservation systems, and government offices in the 1960s–80s. Today, they are rarely used, but the concept lives on in some server environments.

Smart Terminal

A Smart Terminal has a limited amount of its own processing capability — a small processor and some memory. This allows it to do basic tasks locally — like checking that a form has been filled in correctly

before sending it to the central computer, or formatting data for display. This reduces the load on the central computer. However, a smart terminal still depends on the central computer for major processing tasks.

Intelligent Terminal

An Intelligent Terminal is essentially a full personal computer — it has its own powerful processor, its own memory (RAM), its own storage, and its own operating system. It can function completely independently even without a connection to the central computer. It also has the ability to connect to a central server to access shared resources. Today, the computers and laptops we use are all intelligent terminals — they work on their own and also connect to servers via the internet.

■ Real-Life Example

Think of three types of employees in an office. A dumb terminal is like an office peon who can only carry messages between the manager and clients — they have no decision-making power of their own. A smart terminal is like a junior officer who can handle basic tasks but still needs the manager for important decisions. An intelligent terminal is like a senior officer who can work completely independently, but can also consult the manager when needed.

■ Short Summary

Terminals are devices used to interact with central computers. Dumb Terminals have no processing power of their own — completely dependent on the central computer. Smart Terminals have limited processing — can handle basic local tasks. Intelligent Terminals are full computers that can work independently and also connect to central servers.

19. Dedicated Data Entry and Source Data Automation (SDA)

■ Definition

Dedicated Data Entry refers to a system or team set up specifically and solely for the purpose of entering large volumes of data into a computer — it is their only job. Source Data Automation (SDA) is a method of input where data is captured automatically at the point where it is created, without any human typing — using devices like barcode scanners, card readers, or OCR machines.

■ Detailed Explanation

Dedicated Data Entry

In many large organisations — such as insurance companies, tax offices, or hospitals — enormous amounts of data must be entered into computer systems every day. Rather than asking every employee to also do their own data entry (which would slow them down and increase errors), some organisations create dedicated data entry departments. These departments employ trained typists whose only job is to type data from paper forms or source documents into the computer system as accurately and quickly as possible. Dedicated data entry staff use specialised keyboards, are trained for speed and accuracy, and work in structured shifts to handle large volumes.

Source Data Automation (SDA)

SDA is the more modern and efficient approach. Instead of having humans manually type data, SDA devices automatically capture data at the very point where it is created — eliminating manual entry entirely. This dramatically reduces errors and speeds up the entire data collection process.

Common SDA technologies include: Barcode Scanners — a beam of light reads the pattern of bars on a product label and instantly converts it to product information (used in every supermarket checkout and warehouse). Magnetic Stripe Readers — read data stored in the magnetic stripe on the back of credit and debit cards (used at ATMs and point-of-sale terminals). OCR (Optical Character Recognition) — machines that scan printed or typed text and automatically convert it into editable digital text (used in digitising books and reading printed forms). OMR (Optical Mark Recognition) — reads pencil marks made in boxes on forms, such as multiple-choice exam answer sheets (used for grading standardised tests). MICR (Magnetic Ink Character Recognition) — reads special numbers printed in magnetic ink on bank cheques (used by banks to process millions of cheques automatically). RFID (Radio Frequency Identification) — a small chip broadcasts data wirelessly, read by a scanner without any physical contact (used in modern ID cards, toll collection, and inventory management).

■ Real-Life Example

When you check out at a supermarket, the cashier does not type in each product's name and price manually. They simply pass each item under a barcode scanner. The scanner automatically reads the product code, looks up the price in the database, and adds it to your bill — in under a second per item. This is Source Data Automation making retail checkout fast and error-free.

■ Short Summary

Dedicated Data Entry uses trained staff whose sole job is to enter large volumes of data accurately. Source Data Automation (SDA) eliminates manual entry by automatically capturing data at its source using barcode scanners, magnetic stripe readers, OCR, OMR, MICR, and RFID technology. SDA is faster, cheaper, and far less prone to human error than manual entry.

20. Pointing Devices and Voice Input

■ Definition

Pointing Devices are input devices that allow a user to control the position of a cursor on the screen and interact with graphical elements (icons, buttons, menus) by pointing, clicking, and dragging. Voice Input devices allow a user to give commands or enter data by speaking to the computer, which then converts the spoken words into digital text or commands.

■ Detailed Explanation

Pointing Devices

The Mouse is the most widely used pointing device. Moving the mouse on a flat surface moves the on-screen cursor. Buttons on the mouse allow left-click (select), right-click (open menu), and scroll wheel

(scroll through pages). Modern mice are optical — they use a light sensor on the bottom instead of a rolling ball. A Trackball is like an upside-down mouse — the ball is on top and the user rolls it with their thumb or fingers while the body of the device stays still. Often used in industrial settings and by people with limited desk space. A Touchpad (also called a trackpad) is the rectangular touch-sensitive surface on laptops that you slide your finger across to move the cursor. Tapping replaces clicking. A Touch Screen allows the user to interact directly with what they see on the screen by touching it with their finger or a stylus pen — used on smartphones, tablets, ATM machines, and interactive kiosks. A Light Pen is a pen-shaped device that senses light from the screen and lets the user draw or select items by touching the screen directly — used in design and technical drawing applications. A Joystick is a stick that pivots in all directions and translates that movement into on-screen cursor movement — primarily used for video games and flight simulators.

Voice Input

Voice Input (also called Speech Recognition or Voice Recognition) allows users to control the computer or enter data simply by speaking. A microphone captures the user's voice. Special software analyses the sound waves and converts them into text or commands that the computer can process.

Modern voice input has become extremely accurate. Examples include: Dictation software (like Dragon NaturallySpeaking) that converts spoken words directly into typed text — ideal for people who cannot type or for high-speed dictation. Voice assistants like Apple's Siri, Google Assistant, Amazon's Alexa, and Microsoft's Cortana — which respond to spoken commands and questions. Voice search in web browsers. Voice commands in car navigation systems.

Voice input is especially valuable for people with physical disabilities that make typing difficult, for hands-free operation (like driving), and for fast data capture in fields like medicine (doctors dictating notes).

■ Real-Life Example

Think about how you use your smartphone. When you type a message, the touchscreen keyboard is a pointing device responding to your finger touches. When you say 'Hey Siri, set an alarm for 7 AM,' that is voice input — the microphone captures your voice, the speech recognition software converts it to a command, and the phone sets the alarm. Both methods give input to the computer, just in different ways.

■ Short Summary

Pointing devices control the on-screen cursor and include: Mouse (most common), Trackball, Touchpad (on laptops), Touchscreen (direct finger input), Light Pen, and Joystick (games). Voice Input uses a microphone and speech recognition software to convert spoken words into commands or text — used in voice assistants, dictation software, and hands-free systems.

21. Output Devices — Soft Copies and Hard Copies

■ Definition

Output devices are hardware components that deliver the results of computer processing to the user. The two main forms of output are Soft Copies — output that is digital and displayed on a screen or played through speakers, which disappears when turned off — and Hard Copies — output that is printed on paper or another physical medium, creating a permanent tangible record.

■ Detailed Explanation

Once the CPU has processed data, the results must be delivered to the user in a usable form. Output devices are the tools that do this job. The type of output needed depends on the task.

Soft Copy Output

Soft copy output is output in digital, electronic form. It can be seen or heard but cannot be physically held. It exists only as long as the device displaying it is powered on. The most common soft copy output devices are monitors (screens) and speakers. Other examples include projectors (which display digital images on a large screen or wall) and headphones. Soft copy is ideal when you need to view information temporarily, make changes, or share it digitally. Documents displayed on screen, videos streamed online, and music played through speakers are all soft copy outputs.

Hard Copy Output

Hard copy output is output printed or reproduced on a physical medium — typically paper, but also film, fabric, or other materials. Hard copy creates a permanent, tangible record that exists independently of any computer or power source. The most common hard copy output device is a printer. Plotters (which draw precise lines and shapes) also produce hard copies. Hard copy is essential when a physical document is required — for contracts, certificates, reports, books, and official records.

When deciding between soft and hard copy, consider: Does the document need to be physically signed? Hard copy. Will it be shared electronically? Soft copy. Does a permanent physical record need to be kept? Hard copy. Is it a temporary working draft? Soft copy.

■ Real-Life Example

When your teacher shows you a presentation on the projector during a lecture, that is a soft copy — it exists only while the projector is on. When you print your assignment and submit it to your teacher, that is a hard copy — it is a physical document that exists permanently. Both are outputs of the computer, but in different forms.

■ Short Summary

Output devices deliver processed results to the user. Soft copy is digital output — displayed on screens or played through speakers, temporary, not physically tangible (examples: monitor, speakers, projector). Hard copy is physical, printed output — permanent and tangible (examples: printed documents from printers and plotters).

22. Monitors and their Types

■ Definition

A monitor (also called a display or screen) is the primary visual output device of a computer. It displays text, images, videos, and graphical interfaces to the user. Without a monitor, you would have no way to see what the computer is doing. Monitors come in several different technologies, each with its own characteristics.

■ Detailed Explanation

CRT Monitor — Cathode Ray Tube

The CRT monitor was the standard for many decades, from the early days of computing up to the early 2000s. It uses a large glass vacuum tube in which an electron gun fires beams of electrons at a phosphor-coated screen, causing it to glow and produce images. CRT monitors are large, heavy, deep (box-shaped), and consume a lot of electricity. They produce visible flickering if the refresh rate is too low. You rarely see them today — they have been almost completely replaced by flat-panel displays. However, some professionals still prefer CRT monitors for colour accuracy.

LCD Monitor — Liquid Crystal Display

LCD monitors are the flat, thin screens that replaced CRTs and are now used in almost all laptops, desktop monitors, and televisions. They work by shining a backlight (usually fluorescent or LED) through a layer of liquid crystals that can be electronically controlled to block or allow light, creating the image. LCD monitors are much thinner, lighter, and more energy-efficient than CRTs. They produce no harmful radiation and have very sharp, crisp images. Most modern monitors and laptop screens are LCDs with LED backlighting (called LED monitors).

LED Monitor

An LED monitor is essentially an LCD monitor that uses Light Emitting Diodes (LEDs) as the backlight instead of fluorescent tubes. LED backlighting produces brighter, more vivid colours, better contrast, and uses even less power than standard LCD. LEDs can also be used in OLED displays, where each individual pixel produces its own light — resulting in perfect black levels and outstanding colour accuracy. OLED technology is now found in high-end smartphones and premium televisions.

Touchscreen Monitor

A touchscreen monitor responds to touch input from a finger or stylus directly on the screen surface. It combines the functions of an output device (displaying information) and an input device (receiving touch commands) in one unit. Widely used in smartphones, tablets, ATMs, self-service kiosks, and interactive whiteboards in schools.

Monitor Specifications

Key features to consider when choosing a monitor: Screen Size (measured diagonally in inches — e.g., 24", 27", 32"). Resolution (the number of pixels — e.g., 1920x1080 Full HD, 3840x2160 4K — more pixels = sharper image). Refresh Rate (how many times per second the image is updated — measured in Hz;

60Hz is standard, 144Hz is preferred for gaming). Aspect Ratio (the width-to-height ratio — 16:9 is the standard widescreen format).

■ Real-Life Example

Compare an old desktop computer from the year 2000 with a modern one. The old one had a huge, heavy, box-shaped CRT monitor that sat on the desk and took up enormous space. Today's computer has a thin, light, flat LED LCD screen that hangs on the wall or sits on a slim stand. Same function — display output — but dramatically different technology, size, weight, and power consumption.

■ Short Summary

Monitors display visual output from the computer. Types: CRT (old, large, heavy, uses electron gun — now obsolete), LCD (thin, flat, energy-efficient, most common today), LED (improved LCD with LED backlight — brighter and more efficient), OLED (each pixel self-lit — perfect blacks, found in premium devices), and Touchscreen (combined output and input). **Key specs:** screen size, resolution, refresh rate.

23. Printers, their Types, and Plotters

■ Definition

A printer is an output device that produces hard copy — it transfers digital information from the computer onto paper (or other physical media) in the form of text, images, or both. A plotter is a specialised type of output device that draws precise lines and shapes — used for technical drawings, engineering blueprints, and large-scale graphics.

■ Detailed Explanation

Dot Matrix Printer

The dot matrix printer is one of the oldest printer types still occasionally in use. It works by striking an ink-soaked ribbon with a print head containing a matrix of small metal pins. The pins form patterns of tiny dots that together make up characters and simple images. Dot matrix printers are very slow, quite noisy, and produce low-quality output compared to modern printers. However, they are still used in environments that require carbon copies (like banks printing receipts on multi-part paper) because the impact mechanism can print through multiple layers simultaneously.

Inkjet Printer

Inkjet printers are the most commonly used printers in homes and small offices. They work by spraying tiny, precise droplets of liquid ink directly onto the paper through microscopic nozzles in the print head. Inkjet printers can produce high-quality colour output — including photographs — at a reasonable cost. They are quiet, relatively affordable to purchase, and compact. However, the ink cartridges can be expensive to replace frequently, and the ink can smudge if the paper gets wet.

Laser Printer

Laser printers work using a completely different technology. They use a laser beam to trace the image onto a rotating drum coated with a light-sensitive material. The laser changes the electrical charge on parts of the drum, which then attracts fine powder ink called toner. The toner is transferred to the paper and fused permanently by heat. Laser printers are much faster than inkjet printers, produce sharper text, and the toner is cheaper per page than ink cartridges. They are preferred in offices and businesses where large volumes of documents need to be printed quickly. Colour laser printers also exist but are more expensive.

Thermal Printer

Thermal printers produce output by applying heat to special heat-sensitive paper, which turns dark where heat is applied. No ink or toner is required. They are very fast, quiet, and require minimal maintenance. They are widely used for printing receipts in shops, ATMs, and restaurants, and for barcode labels in warehouses and shipping.

Plotters

A plotter is a specialised output device that produces large, precise, high-quality drawings rather than regular text documents. Pen plotters use actual pens to draw continuous lines on paper — producing perfectly smooth curves and lines. Cutting plotters use blades instead of pens to cut shapes out of vinyl or other materials — used for signs and stickers. Large-format inkjet plotters spray ink to produce very large colour prints — used for posters, banners, and technical drawings. Plotters are used by engineers, architects, and designers to print blueprints, schematics, and large-scale maps that cannot be printed on standard printers.

■ Real-Life Example

When your local pharmacy prints your prescription receipt, that is a thermal printer. When you print a family photo at home, that is an inkjet printer. When a law firm prints thousands of pages of case documents, that is a laser printer — chosen for speed and cost efficiency. When a civil engineering firm prints a full-scale building floor plan on a large sheet, that is a plotter at work.

■ Short Summary

Printers produce hard copy output. Types: Dot Matrix (impact pins, slow, noisy, used for carbon copies), Inkjet (sprays liquid ink, good colour quality, common in homes), Laser (uses laser and toner powder, fast and sharp, preferred in offices), Thermal (heat-sensitive paper, no ink, used for receipts). Plotters draw precise large-scale technical drawings — used by engineers and architects.

24. Computer Viruses and their Forms

■ Definition

A computer virus is a malicious (harmful) program written with the intention of damaging, disrupting, or gaining unauthorised access to a computer system. Just like a biological virus infects a human body and spreads from person to person, a computer virus attaches itself to legitimate files or programs and spreads from one computer to another — often without the user's knowledge.

■ Detailed Explanation

When people talk about computer viruses in everyday conversation, they often use the term loosely to mean any harmful software. In technical terms, however, there are many different types of malicious software — collectively called Malware (short for Malicious Software). Each type works differently and causes different kinds of harm.

Virus

A true computer virus is a piece of code that attaches itself to a legitimate program or file. When that infected program is run or that file is opened, the virus activates — it may delete files, corrupt data, display unwanted messages, or slow the computer down significantly. It then copies itself into other programs and files, spreading further. Viruses are typically spread through infected email attachments, downloaded files, or infected storage devices (USB drives).

Worm

A worm is similar to a virus but with one critical difference — a worm does not need to attach itself to an existing file. It is an independent program that can copy and spread itself automatically across a network without any human action. A single worm can replicate thousands of times very quickly, consuming network bandwidth and system resources, causing networks and computers to slow down or crash completely.

Trojan Horse

A Trojan Horse (or simply 'Trojan') is a program that appears to be a useful, legitimate application — like a free game, a utility tool, or a software update — but secretly contains harmful code. Unlike viruses and worms, a Trojan does not self-replicate. Instead, it tricks the user into installing it willingly. Once installed, it may steal passwords, give hackers remote access to the computer, or download additional malware.

Spyware and Adware

Spyware secretly monitors the user's activities — recording keystrokes, capturing passwords, reading emails, and sending this information to a third party without the user's knowledge or consent. Adware automatically displays unwanted advertisements — often pop-up windows — on the user's screen, slowing down the computer and degrading the user's experience.

Ransomware

Ransomware is one of the most dangerous modern malware types. It encrypts all the victim's files — making them completely unreadable and inaccessible — and then demands a ransom payment (usually in cryptocurrency) in exchange for the decryption key. Hospitals, government agencies, and companies have

been devastated by ransomware attacks.

Protection Against Viruses

The main tools for protection include: Antivirus Software (regularly scans for and removes known malware — must be kept up to date), Firewalls (monitor and control network traffic, blocking unauthorised connections), regular Software Updates (fixing known security vulnerabilities), and safe user behaviour (not opening suspicious email attachments, not downloading from unknown sources, not clicking unknown links).

■ Real-Life Example

Imagine you receive an email with an attachment labelled 'Free Movie Download.' You excitedly download and open it. Unknown to you, a virus has just been installed on your computer. It quietly copies itself to all your other files, slows your computer to a crawl, and then sends itself to everyone in your email contact list — spreading to their computers as well. This is exactly how a computer virus spreads in real life.

■ Short Summary

A computer virus is harmful software that damages or disrupts computer systems. Forms include: Viruses (attach to files, spread when files are opened), Worms (self-spreading across networks, no host file needed), Trojans (disguised as useful software, contains hidden malware), Spyware (monitors user secretly), Adware (unwanted advertising), and Ransomware (encrypts files, demands payment). Protection: antivirus software, firewalls, regular updates, and cautious behaviour.

25. Storage Units — Primary and Secondary Memory

■ Definition

Storage Units are the components of a computer system that hold data and programs — either temporarily while the computer is working (Primary Memory) or permanently for long-term use (Secondary Memory). Every piece of data the computer works with — from the operating system to your photos and documents — must be stored somewhere.

■ Detailed Explanation

Just as a human being needs both short-term memory (what you are thinking about right now) and long-term memory (knowledge stored over years), a computer needs both primary memory (fast, temporary) and secondary memory (slower, permanent).

Primary Memory

Primary Memory (also called Main Memory) is directly accessible by the CPU and holds the data and programs that are currently in use. It is the computer's active workspace. Primary memory is very fast but limited in capacity. RAM (Random Access Memory) is the main type — it holds everything the CPU is currently working on. When you open a program, it is copied from the hard drive into RAM so the CPU can

access it quickly. RAM is volatile — its contents are lost when power is cut. ROM (Read-Only Memory) holds the permanent startup instructions (BIOS). Cache Memory is an extremely fast, very small type of memory located on or very close to the CPU chip — it stores frequently used data so the CPU can access it almost instantly without going all the way to RAM.

Secondary Memory

Secondary Memory provides large-capacity, non-volatile (permanent) storage. Data in secondary memory is not lost when the computer is turned off. It is slower to access than primary memory, but far larger in capacity and far cheaper per unit of storage. Secondary memory includes Hard Disk Drives (HDD), Solid State Drives (SSD), USB flash drives, optical discs (CDs, DVDs, Blu-rays), and magnetic tapes. When you save a file, it moves from RAM (primary memory) to secondary storage. When you open a file, it is copied from secondary storage back into RAM.

Memory Hierarchy

Computer memory can be visualised as a hierarchy, from fastest and most expensive at the top to slowest and cheapest at the bottom: Cache Memory (fastest, smallest, most expensive) → RAM → Hard Drive/SSD → Optical Discs → Magnetic Tape (slowest, largest, cheapest). The computer automatically manages this hierarchy to balance speed and cost.

■ Real-Life Example

Think of a chef in a kitchen. Their short-term working memory — the ingredients they are currently using and the recipe steps they are following — is like RAM (fast, immediately accessible, but forgotten when they finish cooking). Their cookbook on the shelf is like secondary storage — it holds all the recipes permanently and can be consulted whenever needed. Their speed of recalling their three favourite recipes they use every day is like cache — super-fast recall of the most frequently needed information.

■ Short Summary

Primary Memory (RAM, ROM, Cache) is fast, directly accessed by the CPU, but limited and volatile (contents lost when power is off). Secondary Memory (hard drives, SSDs, USB drives, optical discs) is permanent, large capacity, and non-volatile. The memory hierarchy from fastest to slowest: Cache → RAM → Hard Drive → Optical Disc → Magnetic Tape.

26. RAM Types, Cache, Hard Disks, Diskettes, and RAID

■ Definition

This topic covers the specific technologies used in computer storage: the different types of RAM (the computer's active working memory), Cache (ultra-fast temporary storage), Hard Disk Drives (the main permanent storage), Diskettes (an older portable storage medium), and RAID (a system for combining multiple hard drives for speed and safety).

■ Detailed Explanation

Types of RAM

DRAM (Dynamic RAM) is the most common type of RAM. It stores data as electrical charges in capacitors, which must be continuously refreshed thousands of times per second to maintain the data — hence the word 'dynamic.' It is slower than SRAM but much cheaper and available in much larger capacities. SRAM (Static RAM) does not need to be refreshed — it uses flip-flop circuits to hold data as long as power is applied. SRAM is significantly faster than DRAM and is used for cache memory, but it is much more expensive and uses more power. SDRAM (Synchronous DRAM) is synchronised with the CPU clock, allowing it to transfer data in perfect sync with the CPU — significantly faster than older asynchronous DRAM. DDR SDRAM (Double Data Rate SDRAM) transfers data on both the rising and falling edge of each clock cycle, effectively doubling the data transfer rate. Modern computers use DDR4 or DDR5, which are very fast and energy-efficient.

Cache Memory

Cache is a very small, extremely fast type of memory built into or very close to the CPU chip. Its purpose is to store copies of the data and instructions the CPU uses most frequently, so the CPU can retrieve them almost instantaneously without waiting for the slower RAM. Caches are organised in levels: L1 cache (smallest and fastest, built directly into the CPU core), L2 cache (slightly larger and slightly slower, still very fast), and L3 cache (larger, shared among multiple CPU cores, still faster than RAM).

Hard Disk Drive (HDD)

A Hard Disk Drive is the most common form of secondary storage. Inside the sealed metal casing are one or more circular magnetic platters that spin at very high speed (typically 5,400 or 7,200 revolutions per minute). A read/write head — similar to the needle of a record player — moves rapidly across the spinning platter surface and magnetises tiny spots to store data (write) or detects the magnetic pattern to retrieve data (read). HDDs can store very large amounts of data (1TB, 2TB, 4TB, or more) at a low cost per gigabyte, but they are slower than SSDs and can be damaged by physical shocks.

Diskettes (Floppy Disks)

Diskettes, commonly called floppy disks, were a very popular portable storage medium from the 1970s through to the late 1990s. They were thin, flexible magnetic disks enclosed in a protective plastic casing — originally 8 inches, then 5.25 inches, and finally 3.5 inches. A standard 3.5-inch floppy disk could store 1.44 MB — enough for simple text documents but laughably small by today's standards. Floppy disks were used to install software, transfer files between computers, and back up important data. They have been completely replaced by USB flash drives and cloud storage.

RAID — Redundant Array of Independent Disks

RAID is a technology that combines multiple hard disk drives to work together as one unified storage system. There are different RAID levels, each with different goals. RAID 0 (Striping) splits data across multiple drives simultaneously — dramatically increasing speed, but offering no protection if one drive fails (all data is lost). RAID 1 (Mirroring) copies identical data to two or more drives simultaneously — if one drive fails, the data is safe on the mirror. RAID 5 combines striping (speed) with distributed parity (fault

tolerance) — a good balance of performance and protection used in servers and data centres. RAID is used by businesses and data centres where data safety and availability are critical.

■ Real-Life Example

Think of RAID as driving two cars carrying the same important cargo (RAID 1). If one car breaks down, the cargo is still safe in the other car. RAID 0 is like splitting the cargo between two cars and driving in convoy — you get there twice as fast, but if either car has an accident, half the cargo is lost. For irreplaceable data, RAID 1's safety is worth the extra cost of a duplicate drive.

■ Short Summary

RAM types: DRAM (common, needs refreshing), SRAM (fast, used for cache), SDRAM and DDR4/DDR5 (modern fast RAM). **Cache:** ultra-fast CPU-side memory (L1, L2, L3 levels). **Hard Disk:** magnetic spinning platters for permanent storage — large capacity, affordable, slower. **Diskettes:** obsolete small portable magnetic storage (1.44 MB). **RAID:** combines multiple hard drives for speed (RAID 0), safety (RAID 1), or both (RAID 5).

27. Optical Disk Storage and Magnetic Tape Backup

■ Definition

Optical Disk Storage uses laser light to read and write data on disc surfaces. Common examples include CDs (Compact Discs), DVDs (Digital Versatile Discs), and Blu-ray discs. Magnetic Tape is a long strip of magnetic material wound on a reel, used for storing very large quantities of data — primarily for backup purposes.

■ Detailed Explanation

CD-ROM — Compact Disc Read-Only Memory

A CD-ROM can store up to approximately 700 MB of data. The 'ROM' part means the data is written once by the manufacturer and can only be read by the user — it cannot be changed or deleted. CD-ROMs were the standard medium for distributing software, music albums, and encyclopaedias through the 1990s and 2000s. A CD-R (Recordable) disc can be written to once by the user. A CD-RW (Rewritable) disc can be erased and rewritten multiple times.

DVD — Digital Versatile Disc

DVDs look identical to CDs but can store much more data — a standard single-layer DVD holds 4.7 GB, while a dual-layer DVD holds up to 8.5 GB. DVDs replaced CDs for distributing films, software, and large games. Like CDs, they come in ROM (read-only), R (write-once), and RW (rewritable) versions. DVDs offered enough capacity to store full-length movies in high quality — making them the dominant home video format from the late 1990s until streaming services took over.

Blu-ray Disc

Blu-ray uses a blue-violet laser (which has a shorter wavelength than the red laser used in DVDs), allowing data to be packed much more tightly. A single-layer Blu-ray disc holds 25 GB, and a dual-layer holds 50 GB. Blu-ray was designed for high-definition (HD) and ultra-HD (4K) video and is still used for premium physical media distribution.

Magnetic Tape

Magnetic tape is one of the oldest data storage technologies — it was used in the very first computers. It consists of a long, thin strip of plastic coated with a magnetic material, wound onto a reel or stored in a cassette cartridge. Data is written and read sequentially — to access data in the middle, the tape must be wound forward past everything before it (like fast-forwarding a cassette tape). This makes magnetic tape slow for random access but it is still the most cost-effective technology for storing enormous quantities of data that is rarely accessed — such as archives, backups, and long-term records. Modern tape cartridges can hold several terabytes of data at a very low cost per gigabyte. Large companies, banks, and cloud providers still use magnetic tape for their deepest archive storage.

Backup Systems

A Backup System is any combination of hardware and software that creates copies of data in a separate location so that if the original data is lost, damaged, or corrupted, it can be restored from the backup. Regular backups are essential for any individual or organisation. Backup strategies include: Full Backup (copies everything — slow but complete), Incremental Backup (copies only files changed since the last backup — fast but requires all incrementals to restore), and Differential Backup (copies all changes since the last full backup — a compromise). Backup destinations include external hard drives, network storage, magnetic tape, and cloud storage (uploading data to secure internet servers).

■ Real-Life Example

Think of optical discs like books in a library. A CD holds a short story (700 MB). A DVD holds a full novel (4.7 GB). A Blu-ray holds an entire encyclopaedia volume (25 GB). Magnetic tape is like a very long filing cabinet in a warehouse basement — you cannot quickly jump to a random item (sequential access only), but you can store an enormous amount at very low cost, making it perfect for archiving data you rarely need to access.

■ Short Summary

Optical storage: CD (700 MB, music/software), DVD (4.7–8.5 GB, films/software), Blu-ray (25–50 GB, HD video). **Magnetic Tape:** sequential access, enormous capacity, very low cost — used for backups and archives. **Backup Systems** create copies of data to protect against loss — strategies include Full, Incremental, and Differential backups. **Cloud storage** is the modern backup method.

28. Data Communications — Model, Transmission, and Modems

■ Definition

Data Communication is the transfer of digital data from one device to another through a transmission medium (cable or wireless signal). The Data Communication Model describes the components involved in any communication. Data Transmission refers to how data travels between sender and receiver. A Modem is the device that makes it possible to send digital computer data over traditional telephone lines.

■ Detailed Explanation

The Data Communication Model

Every data communication involves five basic components: The Sender is the device or person that originates the message (e.g., your computer, your smartphone). The Message is the actual data being sent (text, image, video, file). The Transmission Medium is the physical path through which the message travels (cable, optical fibre, wireless radio waves). The Receiver is the device or person at the other end that receives the message (e.g., a web server, another person's phone). The Protocol is the set of rules and standards that both sides must follow to understand each other (e.g., TCP/IP, HTTP). All five elements must work correctly for communication to succeed.

Analog vs Digital Transmission

Analog transmission sends data as a continuous wave signal — like the continuous variation of sound waves. Traditional telephone lines carried analog signals. Analog signals degrade over distance (noise is added to the signal, distorting it). Digital transmission sends data as discrete binary signals — a series of 0s and 1s represented as high and low voltage levels. Digital signals can be perfectly regenerated and corrected over long distances. Modern computers and networks use digital transmission exclusively.

Modems

Modem stands for Modulator-Demodulator. It is a device that converts (modulates) digital signals from a computer into analog signals that can travel over telephone lines, and converts (demodulates) incoming analog signals back into digital signals the computer can understand. When the internet first became widely available in the 1990s, most homes connected using dial-up modems over telephone lines — a slow and now obsolete technology. Modern cable modems and DSL modems connect homes to broadband internet at much higher speeds. The modem is what bridges the gap between the digital world of computers and the analog world of telephone infrastructure.

Asynchronous vs Synchronous Transmission

In Asynchronous Transmission, data is sent one character at a time. Each character is preceded by a start bit (to signal the beginning of a character) and followed by a stop bit (to signal the end). There is no requirement for sender and receiver to be synchronised — data can be sent at irregular intervals. This is simple and inexpensive, used for low-speed applications like serial ports and older modems. In Synchronous Transmission, data is sent as a continuous stream of characters in blocks, with sender and receiver perfectly synchronised using a common clock signal. There are no start/stop bits for each character — instead, special synchronisation characters mark the beginning and end of each block.

Synchronous transmission is faster and more efficient, used for high-speed network communications.

Simplex, Half Duplex, and Full Duplex

Simplex transmission means data can only flow in one direction — from sender to receiver — never the other way. Example: a television broadcast sends signals to your TV, but your TV cannot send signals back to the broadcaster. Half Duplex means data can flow in both directions, but not at the same time — only one side can transmit at a time. Example: a walkie-talkie — you press the button to speak, release it to listen; both people cannot talk simultaneously. Full Duplex means data can flow in both directions simultaneously — both sides can send and receive at the same time. Example: a telephone call — both people can speak and hear at the same time. Almost all modern computer networks use full duplex communication.

■ Real-Life Example

Think of a conversation between two people. If one person talks through a megaphone to a crowd and the crowd cannot respond — that is Simplex. If two people on walkie-talkies take turns speaking — that is Half Duplex. If two people have a normal telephone conversation speaking and listening at the same time — that is Full Duplex. Data communication works the same way.

■ Short Summary

Data Communication transfers data between devices using a model of: Sender → Message → Medium → Receiver (governed by Protocols). Digital transmission is used by computers (reliable, no degradation). Analog transmission used traditional phone lines (degrades with distance). Modems convert between digital and analog. Asynchronous transmission sends one character at a time; Synchronous sends blocks in sync. Simplex: one-way only; Half Duplex: both ways but not simultaneously; Full Duplex: both ways simultaneously.

29. Communication Media, Protocols, and Network Topologies

■ Definition

Communication Media are the physical or wireless channels through which data travels between devices. Protocols are the agreed-upon rules and standards that define how data is formatted, transmitted, and received. Network Topologies describe the physical or logical arrangement (layout) of computers and devices in a network.

■ Detailed Explanation

Communication Media — Cables

Twisted Pair Cable consists of pairs of copper wires twisted together to reduce electrical interference from external sources. It is the most common and affordable network cable. Used for telephone lines and standard Ethernet (LAN) connections. Available in different categories: Cat5e (up to 1 Gbps), Cat6 (up to 10 Gbps). Coaxial Cable has a central copper conductor surrounded by insulation, a metallic shield, and an outer plastic jacket. It can carry more data than twisted pair over longer distances. Used for cable

television and older network setups. Fibre Optic Cable transmits data as pulses of light through ultra-thin glass or plastic fibres rather than as electrical signals. It is extremely fast (can carry data at the speed of light), not affected by electromagnetic interference, and can transmit over very long distances without signal degradation. Fibre optic is the backbone of the internet and long-distance telecommunications.

Communication Media — Wireless

Wireless media transmit data through electromagnetic waves — no physical cable is required. Radio Waves are used for Wi-Fi networks, Bluetooth (short-range wireless for keyboards, headphones, and peripherals), and mobile phone networks (3G, 4G, 5G). Microwaves are used for point-to-point communication between buildings and for satellite communication — they require clear line of sight between sender and receiver. Infrared is used for very short-range communication — like your TV remote control.

Protocols

A Protocol is a set of rules and standards that defines exactly how data should be formatted, addressed, transmitted, received, and acknowledged between two or more devices. Without agreed protocols, different computers and devices could not communicate — like two people trying to have a conversation in different languages. TCP/IP (Transmission Control Protocol/Internet Protocol) is the fundamental protocol suite of the internet — it defines how data is broken into packets, addressed, routed, and reassembled. HTTP (HyperText Transfer Protocol) is used for transmitting web pages from servers to browsers. HTTPS is the secure, encrypted version. FTP (File Transfer Protocol) is used for transferring files between computers. SMTP is used for sending emails. DNS (Domain Name System) translates human-friendly website names (like www.google.com) into numerical IP addresses that routers can use.

Network Topologies

Star Topology: All devices in the network are connected to a central hub or switch. Every device communicates with every other device through this central point. Advantages: easy to add new devices, easy to troubleshoot (if one device fails, only that device is affected). Disadvantage: if the central hub fails, the entire network goes down. This is the most widely used topology in modern LANs. **Bus Topology:** All devices are connected to a single long cable called the 'bus' or 'backbone.' Data travels along this cable and every device can see it, but only the intended recipient accepts it. Advantages: simple and cheap to set up. Disadvantages: if the main cable breaks, the entire network fails; performance degrades as more devices are added. **Ring Topology:** Each device is connected to exactly two other devices, forming a closed circle (ring). Data travels in one direction around the ring from device to device until it reaches its destination. Advantages: equal access for all devices, no central point of failure. Disadvantages: if any single device or connection in the ring fails, the entire network is disrupted; adding new devices requires breaking the ring.

■ Real-Life Example

Think of network topologies as road layouts. Star topology is like a roundabout — all roads lead to a central junction; if the roundabout is closed, all roads are cut off. Bus topology is like a single long

highway with houses along both sides — if the highway is blocked, everyone is stuck. Ring topology is like a circular road where traffic moves in one direction — if any section of the road is blocked, the whole circle is broken.

■ Short Summary

Communication Media: Twisted Pair (common, affordable LAN cable), Coaxial (cable TV, older networks), Fibre Optic (fastest, uses light, internet backbone), Wireless (Wi-Fi, Bluetooth, mobile networks). **Protocols:** agreed rules for communication (TCP/IP, HTTP, FTP, SMTP). **Topologies:** Star (central hub, most common), Bus (single cable backbone), Ring (circular chain of devices).

30. LAN, Internet, Browsers, and Common Internet Services

■ Definition

A Local Area Network (LAN) is a network of computers and devices connected within a limited geographical area — such as a home, school, or office building. The Internet is a massive, global network connecting millions of computers worldwide. A Web Browser is software that allows users to access and navigate websites on the internet. Common Internet Services include email, the World Wide Web, file transfer, video calling, and many others.

■ Detailed Explanation

LAN — Local Area Network

A LAN connects computers and devices within a single building or a small group of nearby buildings — like a school campus, a hospital, or a company office. All devices on a LAN can share resources — such as a shared printer, a common file server, or an internet connection. LANs use twisted pair cables, Wi-Fi (wireless LAN), or fibre optic cables. They are typically managed by the organisation that owns them and offer fast, private, and secure communication between connected devices.

A Brief History of the Internet

The internet did not appear suddenly. Its origins go back to 1969, when the United States Department of Defence funded a project called ARPANET (Advanced Research Projects Agency Network). ARPANET was the world's first network to use packet switching — the technique of breaking data into small packets, sending them independently across the network, and reassembling them at the destination. Initially, ARPANET connected only four universities. Over the following decades, the network grew and evolved. In 1991, Tim Berners-Lee invented the World Wide Web (WWW) — a system of interlinked documents (web pages) that could be accessed through the internet using a browser. The WWW transformed the internet from a tool for scientists and the military into something accessible to ordinary people worldwide. Today, the internet connects billions of devices and users across every country on earth.

Internet Service Providers and Online Service Providers

An Internet Service Provider (ISP) is a company that provides individuals and organisations with access to the internet — examples include PTCL, Jazz, Zong, and StormFibre in Pakistan; AT&T; and Comcast in the USA. You pay the ISP a monthly fee, and they connect your home or business to the internet through cables, fibre optic lines, or wireless signals. An Online Service Provider (OSP) is a company that provides specific services and content over the internet — such as Google (search), YouTube (video), Netflix (streaming), Amazon (shopping), and Gmail (email). ISPs give you the road; OSPs provide the destinations you visit on that road.

Web Browsers — Functions and Features

A Web Browser is software that allows you to access, retrieve, and display web pages from the internet. When you type a web address (URL) into the browser, it sends a request to the web server at that address, receives the web page data, and displays it on your screen in a readable, formatted form. Common browsers include Google Chrome, Mozilla Firefox, Microsoft Edge, Apple Safari, and Opera. Key features of browsers include: an Address Bar (where you type the URL), Tabbed Browsing (open multiple websites at once in separate tabs), Bookmarks/Favourites (save frequently visited sites), Navigation Buttons (Back, Forward, Refresh), Private/Incognito Mode (browsing without saving history), and Extensions/Add-ons (extra tools that add functionality to the browser).

Search Engines

A Search Engine is a web-based tool that helps users find information on the internet by searching through billions of web pages. You type keywords into the search box, and the search engine displays a list of relevant web pages ranked by relevance. Google is by far the most widely used search engine, followed by Bing (Microsoft), Yahoo, DuckDuckGo (privacy-focused), and Baidu (most used in China). Search engines work by using software programs called 'web crawlers' or 'spiders' that continuously browse the internet, indexing web pages into a giant database that can be searched instantly.

Common Services Available on the Internet

The World Wide Web (WWW) allows users to access interconnected web pages containing text, images, videos, and links — the most visible part of the internet. Email (Electronic Mail) allows users to send text messages, files, images, and documents to other users anywhere in the world, almost instantly (Gmail, Outlook, Yahoo Mail). File Transfer — using FTP or modern file-sharing services, users can upload and download files between computers worldwide. Online Communication services include instant messaging, video calling (Zoom, Skype, WhatsApp), social media (Facebook, Instagram, Twitter/X), and forums. E-Commerce allows people to buy and sell products and services online (Amazon, Daraz, eBay). Online Banking allows users to view accounts, transfer money, and pay bills through internet banking portals. Cloud Storage services (Google Drive, Dropbox, iCloud) allow users to store files on remote servers and access them from any device, anywhere.

■ Real-Life Example

Imagine the internet as a vast city. Your ISP builds the road from your house to the city (your internet connection). Your web browser is your car — the vehicle you use to drive around and visit places.

Search engines like Google are the city's directory or GPS — they help you find exactly where you want to go. Websites like YouTube, Gmail, and Amazon are the shops, restaurants, and offices in the city — each offering a different service.

■ Short Summary

LAN: local network within a building for sharing resources. Internet: global network of networks, born from ARPANET (1969), transformed by the World Wide Web (1991). ISPs provide internet access (PTCL, Jazz). OSPs provide internet content (Google, Netflix). Browsers (Chrome, Firefox) access and display web pages. Search engines (Google, Bing) help find information online. Common internet services: WWW, Email, File Transfer, Video Calling, E-Commerce, Online Banking, Cloud Storage.

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